

Mohd Arif Ansari

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Portfolio

Career Highlights

AI Engineer with 3.5+ years of experience building **scalable AI systems and data platforms**, combining strong expertise in **Generative AI (RAG, Agentic AI, LLM fine-tuning)** with **production-grade data engineering**. Proven track record of developing **end-to-end ML pipelines**, real-time data systems, and **LLM-powered applications** with measurable improvements in accuracy, latency, and cost. Skilled in bridging **data infrastructure with AI models**, enabling reliable and efficient deployment in real-world environments.

AI/ML & Data Engineering Stack

AI/ML & Generative AI: PyTorch, Scikit-learn, Transformers (BERT, LayoutLM), LLMs (OpenAI, Gemini, LLaMA), Prompt Engineering RAG, Agentic AI (Multi-Agent Systems), Fine-tuning (LoRA, PEFT), Embeddings, Vector Databases, MLflow

Data Engineering & Cloud: Apache Spark, PySpark, Structured Streaming, Delta Lake, Kafka, ETL/ELT Pipelines, Data Modeling, Data Validation, Azure Databricks, ADLS Gen2, Azure Data Factory, Synapse Analytics, PostgreSQL, Snowflake, SQL Server, NoSQL

Programming & Tools: Python (Pandas, NumPy), SQL, FastAPI, Docker, CI/CD (GitHub Actions, Azure DevOps), Power BI

Professional Experience

Data Engineer | Ingenious e-Brain ,06 Jan 2025 – 23 May 2025

- Engineered distributed **PySpark pipelines on Azure Databricks**, processing **10M+ records/week**.
- Built **real-time streaming systems (Kafka + Spark)** handling **5K+ events/min**, reducing latency by **40%**.
- Implemented **Delta Lake architecture** (ACID, schema evolution, time travel) for reliable data processing.
- Designed **low-latency (<500 ms) data platforms** for analytics applications.
- Optimized Spark jobs (partitioning, caching, joins), improving performance by **35%**.
- Automated ETL workflows with validation, improving **data reliability and consistency**.
- Integrated pipelines with **Snowflake, ADLS, CI/CD**, enabling production deployments.

Data Analyst | Provana India , 07 Jan 2021 – 30 Dec 2024

- Built scalable **ETL pipelines using PySpark & Snowflake** for enterprise data systems.
- Developed **data warehouse pipelines**, improving query performance and analytics efficiency.
- Designed **data integration frameworks** across multiple sources.
- Implemented **data quality checks, governance, and validation pipelines**.
- Optimized SQL queries and ETL jobs, improving performance by **~40%**.
- Delivered **ML-ready datasets** by collaborating with data science teams.

Key Projects

Personal AI Chatbot (RAG + Self-RAG System)

- Built an advanced **LLM-powered chatbot using RAG with Self-RAG validation loop**
- Implemented pipeline: ingestion → chunking → embeddings → retrieval → generation → self-correction
- Reduced hallucinations by **~30–35%** using response validation and re-retrieval strategies
- Optimized retrieval using **top-k tuning**, improving relevance by **~25%**
- Enabled **multi-turn conversations with memory**
- Developed modular APIs using **FastAPI (Docker-ready)**

Tech: LangChain, FAISS, LLM APIs, Python

LLM Fine-Tuning & Optimization Pipeline

- Fine-tuned LLMs using **LoRA/PEFT**, improving task performance by **~20–25%**
- Reduced cost via **parameter-efficient fine-tuning** instead of full model retraining
- Optimized inference latency using:
 - Caching mechanisms**
 - Prompt size optimization
 - Async/batch processing
- Built **cost-aware LLM pipeline** with model routing strategies

Tech: HuggingFace, PEFT (LoRA), FastAPI, Python

Computer Vision System (Medical Image Classification)

- Built **CNN/EfficientNet-based model** for image classification (benign vs malignant)

- Achieved ~~~88~~**90**% AUC using transfer learning and augmentation
- Integrated **Grad-CAM explainability** for model interpretability
- Designed full pipeline: preprocessing → training → evaluation

Tech: PyTorch, OpenCV, EfficientNet, Grad-CAM

Certifications: - **Microsoft Certified:** Azure AI Engineer, Data Scientist, Data Engineer, Data Analyst

Education

Master of Technology in AI & ML, Lovely Professional University.	2025 - Present
Postgraduate in Advance Data Science and ML, IIT Madras.	2023 - 2024
Bachelor of Technology in AI & ML, Lovely Professional University.	2016 - 2020